MODULE 02

Statistics and Probability

A concise 10-page starter workbook for Data Science students

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Learning goal

This workbook introduces the core ideas in statistics and probability, with a topic-by-topic study path, practical checkpoints, and reflection prompts. Use it as a starting point and supplement it with your class notes and recommended references.

Note: These are sample learning materials created for this demo portal, not official university course notes.

Population, Samples, and Sampling Methods

Concept overview

In this section, focus on the key principles behind **population, samples, and sampling methods**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain population, samples, and sampling methods to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Descriptive Statistics

Concept overview

In this section, focus on the key principles behind **descriptive statistics**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain descriptive statistics to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Probability Fundamentals

Concept overview

In this section, focus on the key principles behind **probability fundamentals**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain probability fundamentals to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Conditional Probability and Bayes' Theorem

Concept overview

In this section, focus on the key principles behind **conditional probability and bayes' theorem**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain conditional probability and bayes' theorem to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Random Variables and Probability Distributions

Concept overview

In this section, focus on the key principles behind **random variables and probability distributions**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain random variables and probability distributions to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Normal Distribution and Z-Scores

Concept overview

In this section, focus on the key principles behind **normal distribution and z-scores**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain normal distribution and z-scores to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Confidence Intervals

Concept overview

In this section, focus on the key principles behind **confidence intervals**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain confidence intervals to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Hypothesis Testing

Concept overview

In this section, focus on the key principles behind **hypothesis testing**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain hypothesis testing to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Correlation and Statistical Interpretation

Concept overview

In this section, focus on the key principles behind **correlation and statistical interpretation**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain correlation and statistical interpretation to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?